

Bringing Instant Neural Graphics Primitives to Immersive Virtual Reality

Ke Li^{1,3,*}Tim Rolff^{1,2,*}Susanne Schmidt¹Reinhard Bacher³Simone Frintrop²Wim Leemans³Frank Steinicke¹¹ Human-Computer Interaction Group, Department of Informatics, Universität Hamburg² Computer Vision Group, Department of Informatics, Universität Hamburg³ Deutsches Elektronen-Synchrotron DESY, Germany

Figure 1: Illustration of the standard evaluation fox scene from the instant neural graphics primitives framework rendered for a head-mounted display using the Unity engine.

ABSTRACT

Neural radiance field (NeRF), in particular, its extension by instant neural graphics primitives is a novel rendering method for view synthesis that uses real-world images to build photo-realistic immersive virtual scenes. Despite its enormous potential for virtual reality (VR) applications, there is currently little robust integration of NeRF into typical VR systems available for research and benchmarking in the VR community. In this poster paper, we present an extension to instant neural graphics primitives and bring stereoscopic, high-resolution, low-latency, 6-DoF NeRF rendering to the Unity game engine for immersive VR applications.¹

Index Terms: Neural Radiance Field—Instant Neural Graphics Primitives—Immersive Experiences—Free-viewpoint Videos;

1 INTRODUCTION

Advancements in head-mounted displays (HMDs) and graphics processing units (GPUs) allow for immersive virtual reality (VR) experiences, which can realistically display scenes from the real world. Such accurate 3D representations of real-world environments are crucial for various immersive VR applications. However, after acquiring the images of the real-world environments, state-of-the-art 3D reconstruction pipelines require computationally expensive processes based on techniques such as photogrammetry or simultaneous localization and mapping (SLAM).

Recently, *neural radiance field (NeRF)* [1,2] has rapidly emerged as a highly robust 3D reconstruction method for creating photo-realistic 3D scenes based on multiple 2D images and their camera poses only. While the original work by Mildenhall et al. [1] requires

extensive computation for training a single scene, Müller et al. recently proposed *instant neural graphics primitives (instant-ngp)* [2] with a multi-resolution hash encoding, which simultaneously enables rapid training and real-time volumetric rendering. The breakthrough of instant-ngp makes NeRF applicable and of high importance to many VR applications. However, there is currently no robust open-source VR NeRF integration for typical VR game engines for rapid prototyping and benchmarking.

To bridge this current gap in the availability of NeRF software systems, we develop *immersive instant-ngp*, an adaptation of the original instant-ngp framework [2] with integration into Unity, which is one of the most widely used game engines for VR research. Our system simultaneously renders two instances of high-resolution NeRF images, creating a stereoscopic view of the NeRF scene in VR. In addition, *immersive instant-ngp* enables six degrees of freedom (DoF) continuous locomotion, exocentric manipulation of volume image slices of the scene, and is compatible with the Mixed Reality Toolkit (MRTK)² for rapid prototyping of mixed reality (MR) applications based on NeRF. To our best knowledge, this is the first open-source real-time VR-NeRF framework for Unity.

2 IMMERSIVE INSTANT-NGP FRAMEWORK

Framework Overview: Figure 2 gives an overview of the architecture of our framework. It renders a NeRF 3D scene based on a pre-trained instant-ngp model with compressed volumetric scene representation. The Unity application uses input from the six DoF tracking of the VR HMD and its controllers. Based on these transformations, a corresponding view matrix describing the camera rotation and translation is sent to instant-ngp for view synthesis and rendering. Finally, the Unity OpenVR desktop plugin displays the rendered content on the VR HMD.

Image Pre-processing: To generate a neural radiance representation of a 3D scene from a set of unordered 2D images, camera poses need to be estimated. The required transformation matrices

*These authors contributed equally to the work.

¹Link to the repository: <https://github.com/uhhhci/immersive-ngp>

²<https://github.com/microsoft/MixedRealityToolkit-Unity>

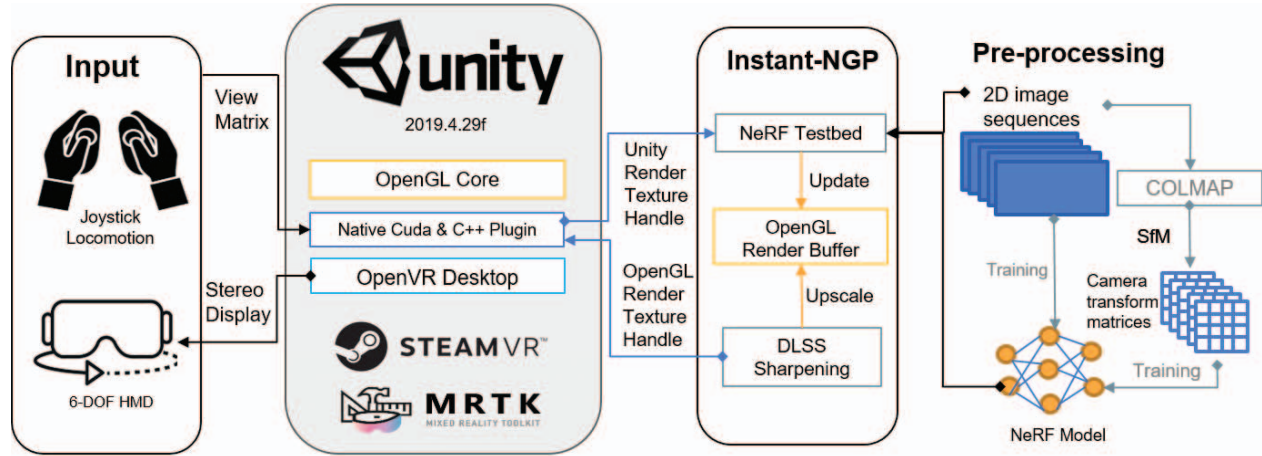


Figure 2: The system architecture of our framework, depicting the individual processes from left to right: Starting with the input, a view matrix is computed from VR input devices and the HMD. The view matrix is sent to the native Unity plugin, which provides a communication layer between instant-ngp and Unity. The instant-ngp backend performs view synthesis through the pre-trained NeRF model and updates two OpenGL textures for stereoscopic rendering. Lastly, the pre-processing refers to the standard instant-ngp model generation process.

are typically computed using conventional photogrammetry tools such as COLMAP [3], which integrates a general-purpose structure from motion (SfM) algorithm.

The Unity Application: The main VR application is driven by Unity with the OpenVR Desktop runtime³. Currently, the original instant-ngp framework [2] creates render context using OpenGL to provide support for both Linux and Windows platforms. Although OpenXR proposed by the Khronos Group is becoming the de facto standard for cross-platform VR⁴, to date, the OpenXR runtime of many popular VR headsets such as the Oculus Quest 2 does not support OpenGL in Unity Windows platforms. To adapt for instant-ngp's dependency on OpenGL, the runtime we use to access different VR hardware is a fallback to OpenVR running together with Steam VR⁵. Therefore, our framework currently runs on Unity version 2019.4.29f1, which is compatible with both the OpenVR desktop and the OpenGL graphics API.

Instant-ngp Native Unity Plugin and Rendering: To support direct communication between Unity and the instant-ngp application, we developed a native Unity C++ plugin, which provides a Unity runtime with interfaces to instant-ngp through pre-compiled dynamic link libraries (DLL). As both the Unity Windows VR application and instant-ngp run on OpenGL, a single render buffer per eye is instantiated at the beginning of the runtime and shared between the Unity application and an instant-ngp instance. To update the render texture in Unity, only OpenGL texture pointer handles need to be shared from the native plugin. To enable immersive stereo vision, two cameras are positioned inside the NeRF scene with a horizontal separation adjustable to any interpupillary distance (IPD) value. Rendering resolution and the field of view can be manually adjusted to accommodate the performance capability of different GPU hardware. Devices with Nvidia DLSS support can enable super-resolution and sampling for better rendering performance.

VR Interactions: To support smooth, uninterrupted movement in VR, our framework implements continuous locomotion, where users could travel by natural walking, or by controller-based navigation for stationary usage. In addition, immersive instant-ngp

also supports rendering and manipulations of volume image slices exported from the instant-ngp with a direct volume ray-casting algorithm adapted from the Unity Volume Rendering package⁶.

Performance: On a Quadro RTX 8000 GPU and at 1280×720 per eye resolution with DLSS support, the current system runs at 15 - 25 frames per second with 54ms average frame time for the standard evaluation fox scene, and 30-40 fps for the standard synthetic lego scene with 27ms average frame time. This suggests that the current system can effectively provide real-time, high-resolution immersive VR experiences for small to medium size NeRF scenes.

3 CONCLUSION AND FUTURE WORK

This poster presents an open-source software framework that brings the 3D reconstruction and rendering power of instant-ngp to immersive VR. With integration into the popular VR game engine Unity, immersive instant-ngp will enable rapid prototyping of VR-NeRF applications in domains such as robot teleoperation, cultural participation, and entertainment. In the future, immersive-ngp will continue to evolve as an open-source project for the VR community, including future releases of foveated NeRF for large-scale scene rendering and immersive 3D photo editing and interaction techniques.

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⁴<https://www.khronos.org/openxr/>

⁵<https://store.steampowered.com/app/250820/SteamVR/>

⁶<https://github.com/mlavik1/UnityVolumeRendering>